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- Oct. 2024 – present: Research Director, Hudson Carbon
- Oct. 2023 – Oct. 2024: Postdoctoral Associate, Yale University School of the Environment
- Oct. 2021 – Oct. 2023: National Science Foundation Postdoctoral Research Fellow in Biology, Yale University School of the Environment

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EDUCATION
- Sept. 2014 – Sept. 2021: UCSB Ecology, Evolution, and Marine Biology (EEMB); PhD in ecology
  - "Biodiversity and ecosystem function: How experimental large herbivore loss influences savanna carbon dynamics"
  - Interdepartmental Emphasis in Environment and Society (IEES)
- Sept. 2014 – June 2018: UCSB EEMB; M.A. in Ecology
- Sept. 2008 – May 2012: Vassar College; Bachelor of Arts with honors in biology

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PUBLICATIONS: **denotes undergraduate author*
1. *In review at Ecosphere:* J. Gewirtzman, A. Keiser, M. Neiland, C. Palmer, V. Patel, K. Caylor, E. Forbes. Low-cost autonomous chambers enable high spatial and temporal resolution monitoring of soil CO_2 emissions across landscapes.
2. *In review at Functional Ecology:* Le Roux, E., ... , Forbes E.S., ... Schoelynck J. Zoogeochemistry: The role of animals in biogeochemical cycles.
3. *In review at Journal of Animal Ecology:* Ferraro, K., Meyer G.A., Roman J., Orr D., Leroux S., Subalusky A., Duvall E.S., Monk J., Gewirtzman J., Rizzuto M., Koltz A., Owens C., Abraham A., Doughty C., Schmitz O.J., Forbes E.S. A practical primer for zoogeochemistry: Essential biogeochemical methods for animal ecologists.
4. Ferraro K., Forbes E., Abraham A., Monk J. 2026. Hot spots or hot moments? Contextualizing the spatio-temporal scale of research on animal inputs. *Ecography*. https://doi.org/10.1002/ecog.08351
5. Ferraro, K., Ferraro A. L., Monk J., Forbes E. 2025. Outdated reproductive norms, the naturalistic fallacy, and misunderstandings of welfare in recent call for zoo breed-and-feed programs. *PNAS* 122 (26) e2504225122. https://doi.org/10.1073/pnas.2504225122
6. Alex McInturff, Peter S. Alagona, Scott D. Cooper, Kaitlyn M. Gaynor, Sarah E. Anderson, Elizabeth S. Forbes, Robert Heilmayr, Elizabeth H.T. Hiroyasu, Bruce E. Kendall, Alexis M. Mychajliw, Molly Hardesty-Moore, Jessica West. 2025. Triangulating habitat suitability for the locally extirpated California grizzly bear. *Biological Conservation*. https://doi.org/10.1016/j.biocon.2025.110989

7. Forbes, E.S., Dunkin Dana D.*, Mantas J.M., Schimel J., Young T.P., Young H.S. January 7, 2025. Large herbivores influence soil carbon cycling as engineers of savanna tree canopy coverage. *Ecosystems* 28(10). https://doi.org/10.1007/s10021-024-00944-7
8. Pan, C., Patel V., Gewirtzman J., Richardson I.*, Dubey R., Caylor K., Dollar A., Forbes E.S. 2024. Fluxbot: The next generation – Design and validation of a wireless, open-source mechatronic CO_2 flux sensing chamber. *Proceedings of the 7th ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies*.
9. Forbes, E. S., Benenati V*, Frey S*, Hirsch M, Koech G, Lewin G*, Mantas NJ, Caylor K. June 2023. Fluxbots: an inexpensive, autonomous soil carbon flux chamber for improving the characterization of landscape-scale fluxes. *Journal of Geophysical Research – Biogeosciences.* 128(6) e2023JG007451. https://doi.org/10.1029/2023JG007451
    a. Editor highlight, *EOS:* "An Open and Inexpensive 'Fluxbot' for Measuring Soil Respiration". July 2023.
    b. Story, *EOS*: "Affordable robots measure soil respiration", December 8, 2023.
10. Miller-ter Kuile, Ana, Apigo A., Bui A., Butner K., Childress J. N., Copeland S., DiFiore B., Forbes E. S., Klope M., Motta C. I., Orr D., Plummer K. A., Preston D. L., Young H. 2022. Changes in invertebrate food web structure between high and low productivity environments are driven by intermediate but not top-predator diet shifts. *Biology Letters* 18: 20220364. https://doi.org/10.1098/rsbl.2022.0364
11. Miller-ter Kuile, Ana, Apigo A., Bui A., DiFiore B., Forbes E. S., Lee M., Orr D., Preston D. L., Behm R., Bogar T., Childress J., Dirzo R., Klope M., Lafferty Kevin D., McLaughlin J., Morse M., Motta C., Park K., Plummer K., Weber D., Young R., Young H. 2022. Predator-prey interactions of terrestrial invertebrates are determined by predator body size and species identity. *Ecology*. https://doi.org/10.1002/ecy.3634
12. Forbes, E. S., Adams A.J., Brown K.C., Colby J.C., Denny S.M., Hardesty-Moore M., Heilmayr R., Hiroyasu E.H.T., Martin J., Miljanich C., Tyrrell B.P., Welch Z., Anderson S.E., Cooper S.D., Kendall B.E., Alagona P.S. 2020. Analogies in a no-analog world: Reintroducing lost species in the Anthropocene. *Trends in Ecology and Evolution: Science and Society*. DOI: https://doi.org/10.1016/j.tree.2020.04.005
13. Forbes, E. S., Cushman J.H., Burkepile D.E., Young T.P., Klope M., Young H.S. 2019. Synthesizing the effects of large, wild herbivore exclusion on ecosystem function. *Functional Ecology*. DOI: 10.1111/1365-2435.13376
14. Lafferty, KD, McLaughlin JP, Gruner DS, Bogar TA*, Bui A, Childress JN, Espinosa M, Forbes ES, Johnston CA, Klope M, Miller-ter Kuile A, Lee M*, Plummer KA, Weber DA*, Young RT, Young HS. 2018. Local extinction of Asian tiger mosquitoes (*Aedes albopictus*) following rat eradication on Palmyra Atoll. *Biology Letters*, 14(2).
15. Forbes, E. S., Miller ter Kuile A., Orr D., Titcomb G. 2016. Navigating the cascades of circumstance: an ecologist reflects on the unexpected twists & turns that shaped his scientific career. *Science.* 352(6289): 1062.

White papers:

1. Bradford, M.A., Oldfield, E.E., Arredondo, M.G., Black, H.I.J., Forbes, E.S., Jevon, F.V., Kelly, C., Lavallee, J.M., Liu, S., Polussa, A., Potash, E., Rosenzweig, S.T., Sanderman, J., Sheban, K.C., Sheffer, M., Treakle, T., Tully, K., Kuebbing, S.E. and the Synthesizing Science Lab (2025) *Agricultural soil carbon: A call for improved evidence of climate mitigation*. Yale Applied Science Synthesis Program and Environmental Defense Fund white paper. https://doi.org/10.31219/osf.io/uk3n2_v1
2. Forbes, E.S., Kordell T., Joyce F., Visalli M., Morse M., McCauley D. (2019) *River Plastic Pollution: Considerations for addressing the leading source of marine debris.* Benioff Ocean Sciences Laboratory. DOI: 10.13140/RG.2.2.20699.80165

Pre-prints and Manuscripts in Preparation:

1. *In preparation:* Forbes E., Arndt K., Sheffer M., Sanderman J. "Organic management results in net uptake of carbon dioxide in a Northeast farm as measured by eddy covariance and survey-based soil respiration".
2. *Preprint, submitted to Global Change Biology:* Polussa A., Oldfield E.E., Runge T., Evans E., Liu S., Forbes E.S., Partida C., Potash E., Sanderman J., Sheffer, Bradford M. "Toward causal designs to quantify management-driven soil carbon change at multi-field scales." https://doi.org/10.31220/agriRxiv.2026.00422

Conference presentations:

1. "The mosaic of the savanna: Exploring rangeland zoogeochemistry to characterize animal impacts at multiple scales". *Inspire session: Carbon on the Range – Improving Our Understanding of Carbon Storage and Dynamics in Rangelands*. ESA 2023.
2. *Poster,* Forbes, Hirsch M., Frey S., Benenati V., Lewin G., Caylor K. "Fluxbot: Using inexpensive sensors and do-it-yourself technology to expand ecologists' collection of, access to, and resolution in *in situ* soil carbon flux measurements". American Geophysical Union 2023.
3. "Biodiversity and ecosystem function: How experimental large herbivore loss influences savanna carbon dynamics. ESA, August 2022.
4. "Wild robots: Developing DIY technology to investigate soil carbon flux in a long-term, landscape-scale, large herbivore exclosure experiment in a central Kenya savanna". ESA 2020.
5. *Poster,* Forbes, Caylor K., Schimel J., Hirsch M., Young H. S., Young T. P. "Animal, vegetable, mineral: Identifying the indirect effects of large vertebrate herbivores and land use on carbon cycling in a Kenyan savanna ecosystem." AGU 2019.
6. "The reintroduction gap: The past, present, and future of the California grizzly bear." Western Section of the Wildlife Society annual meeting. February 8th, 2019.
7. "Connecting above & below: The effects of large herbivore loss & pastoralism on savanna carbon dynamics". ESA 2017.
8. *Poster*, Forbes, Young T., Young H., Schimel J. "Connecting above and below: The impacts of large wildlife loss and pastoralism on savanna carbon dynamics". AGU 2016.
9. *Poster*, Forbes, Cardon Z. "The role of *Spartina Alterniflora* in delivering O_2 deep into salt marsh sediments". ESA 2014.

10. *Poster*, Forbes, Cardon Z. "Planar optodes reveal that light intensity aboveground affects the balance of oxygen supply & demand around *Spartina alterniflora* roots deep in salt marsh sediments". 2014 Plum Island Long-Term Ecological Research site (LTER) conference.
11. *Poster*, Forbes, Cardon Z. "Oxygen Probe: Non-destructive, quantitative measurements of oxygen concentration in salt marsh sediments using a planar optode". 2013 Plum Island LTER conference.

SUCCESSFULLY AWARDED GRANTS, CERTIFICATIONS

- Awarded a travel grant from Portable Photosynthesis Systems for presenting research at 2024 Ecological Society of America conference ($1000)
- Awarded a research grant from the Yale Center for Natural Carbon Capture (Spring 2022; $271,000) (awarded with Drs. Oswald Schmitz, Shawn Leroux)
- Awarded NSF Postdoctoral Research Fellowship in Biology (Fall 2021; $138,000)
- Awarded Schmidt Family Foundation Research Accelerator Award (Spring 2019) ($8,000)
- Awarded the Ellen Schamberg Burley Graduate Scholarship (Fall 2019) ($250)
- Co-authored two NSF Research Experience for Undergraduates (REU) (Summers 2018; 2017) ($8,178; $8,778)
- Awarded NSF Integrative Graduate and Education Research Traineeship (IGERT) innovation grant (Spring 2017) ($12,000)
- Awarded NSF Graduate Research Fellowship (GRF) (Spring 2016) ($68,000)
- Awarded National Geographic Young Explorers grant (September 2015) ($5,000)
- Awarded NSF-IGERT fellowship (Fall 2014) ($68,000)

ECOLOGY AND CONSERVATION WORK EXPERIENCE

- September 1-November 1 2020: Contractor, Natural Resources Defense Council (NRDC). Conducted a literature review and synthesis of marine protected area (MPAs) impacts on conservation and fisheries, which contributed to a policy brief with recommendations geared for federal policy makers
- November 1, 2018-May 31, 2019: Staff scientist, Benioff Ocean Initiative (BOI). Conducted review on riverine contributions to marine plastic pollution, existing technologies/initiatives to solve pollution at major source rivers. Lead author on white paper (see *publications*) put forth by BOI. Developed a multi-million dollar grant opportunity for groups innovating river plastics pollution solutions.
- June 2012-August 2014: Research Assistant, Marine Biological Laboratory. Woods Hole, MA. Research Assistant in the lab of Dr. Zoe Cardon. Lab and field work exploring oxygen release to anoxic soils from salt marsh plant roots; analyses of data describing oxygen distribution and consumption at depth (see *presentations*).
- June-August 2011: NSF-REU at Mt. Desert Island Biological Lab. Salisbury Cove, ME. NSF Research Experience for Undergraduates (REU) grantee advised by Dr. Robert Preston, Illinois State University. Conducted field and lab work on the chemistry of killifish embryos, and applications of these data toward killifish embryo desiccation resistance in the intertidal.

- June-August 2010: Northeastern University Marine Science Center, Nahant, MA.
  Laboratory Technician in Dr. Geoffrey Trussell's lab, working primarily with PhD students on the effects of invasive green crabs on dogwhelk shell thickness, behaviors, and consumption along a climate gradient in New England rocky intertidal zones.

WORKSHOPS AND SERVICE (* indicates organizer of event)

- * May 21st – 22nd, 2025: organized 'Hudson Carbon Science Retreat' for regional scientists, focused on identifying synergistic research, possibilities for collaboration, and funding opportunities for continued agroecology research
- * Aug. 6-11, 2023: co-organized a series of two organized oral session and one symposium on zoogeochemistry at annual Ecological Society of America conference
- * May 1st, 2nd 2019: organizer of a science communication workshop for graduate students in the environmental sciences, featuring Dr. Carlie Weiner (Schmidt Ocean Institute) and four science writers from LA Times, NY Times, and YouTube
- * March 16, 2018: organized graduate student workshop, "Politics and Policy of the Environment" – Identified and invited expert speakers, led workshop. Topics included history of environmental legislation; political processes of policy making; and roles of science and scientists in policy today.
- * April 28, 2017 organized communication workshop (Union of Concerned Scientists)
- * Winter 2016-Spring 2017: Beyond Research Collective – organization to for advancement in communication, advocacy, inclusivity, and environmental policy skills
- September 2014-Spring 2018: UCSB EEMB Graduate Student Advisory Committee
  GSAC secretary for four years; spearheaded departmental fundraising and advanced skills-building workshops

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INTERDISCIPLINARY RESEARCH TRAINING AND EXPERIENCE:

- Project lead and Hudson Carbon liaison for collaborative research projects spanning university, nonprofit, and other partners
  - Cornell University, Cornell Cooperative Extension, Columbia University, Yale University, the Cary Institute, the University of New Hampshire, the University of Vermont, Bigelow Laboratory for Ocean Sciences, Woodwell Climate Research Center, the White Buffalo Land Trust, the Quinn Institute, the Northeast Organic Farming Association – NY, the Center for Agricultural Development and Entrepreneurship, Hawthorne Valley Farmscape Ecology
- Project lead on design, development, and deployment of autonomous, robotic soil carbon dioxide emissions sensors (see *publications* and *presentations*).
- California Grizzly Research Network: founding members of the CGRN, led authorship of an informational framework for species reintroduction
- Interdepartmental PhD Emphasis in Environment and Society
- National Science Foundation Interdisciplinary Graduate Education and Research Traineeship focusing on mathematical study of networks and big data

TEACHING:

- September 2024, ***Instructor***: National Garden Clubs, Inc.: Environment School. “Ecology – Air” and “Plants – The Rain Forest” modules
- Fall 2023, ***Instructor of Record***: “Introduction to Ecology for Environmental Managers: Ecological Pattern and Process” (ENV511), Yale School of the Environment, Yale University
- September 2023, ***Instructor***: National Garden Clubs, Inc.: Environment School. “Ecology – Land” and “Environmental Science” modules
- Fall 2021, Guest Lecturer: “Introduction to Ecology” (EEMB 120), EEMB, UCSB
- Summer session A, 2020 *and* 2021: ***Instructor of Record***, “Introduction to Ecology” (EEMB 120) session A, Ecology Evolution and Marine Biology, UCSB
- Spring 2021: ***Instructor of Record***, “Introduction to Ecology” (EEMB 120), EEMB, UCSB
- Winter 2021: Teaching Assistant, “Food, Agriculture, and the Environment”, Environmental Studies/Geography, UCSB
- Fall 2020: Teaching Assistant, “Vertebrate Biology” (EEMB 113L), EEMB, UCSB
- Winter 2020: Teaching Assistant, “Food, Agriculture, and the Environment”, Environmental Studies/Geography, UCSB
- Fall 2017, Guest Lecturer: “Wildlife in America”, Environmental Studies, UCSB
- Fall 2017: Teaching Assistant, “Wildlife in America”, Environmental Studies, UCSB
- Volunteer with “Skype a Scientist” since 2016. Meet with students (pre-K to high school) over virtual meeting format about science, conservation, art and writing, and careers in environmental science and ecology fields

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MENTORSHIP

- 2024-2025: Co-advisor and mentor to two master’s students with the Yale School of the Environment (YSE) on thesis research pertaining to the effects of agricultural management practices on ecosystem soil health, soil biogeochemistry, and greenhouse gas emission.
- 2021-2024: Mentor to four YSE masters’ students, three YSE PhD students
    - Awarded 2024 “Mentorship Award” by the Yale Office for Postdoctoral Affairs, recognizing nominated and recommended postdoctoral scholars who exemplify the role of mentor to undergraduate students or postgraduates
    - 2021-2024: instituted weekly “mentorship coffee hour”, where senior lab members in the Schmitz lab provide support and advice to junior lab members
- 2023-2024: Mentor to two PhD students in Yale School of Engineering and Applied Sciences
- Fall 2023-spring 2024: Primary mentor and lab supervisor for Yale undergraduate research assistant on soil sample processing and basic lab skills
- Summer 2023: Primary mentor and field supervisor for Memorial University undergraduate research assistant on summer research expedition to Newfoundland, Canada
- Summer 2019: Primary mentor for undergraduate student researcher at Mpala Research Centre (Laikipia, Kenya) summer field expedition

- Winter-summer, 2019: Primary mentor to two undergraduate student researchers (physics/engineering; computer science) developing novel tool to measure soil carbon flux
- Summer 2018: Primary mentor for NSF-REU student at Mpala Research Centre (Laikipia, Kenya) summer field expedition
- Summer 2017: Primary mentor for NSF-REU student at UCSB researching the impacts of wildlife community composition on carbon, nitrogen, and microbial biomass of soil samples
- Aug. 2017, 2020: Strategies for Education in Ecology, Diversity, and Sustainability (SEEDS) mentor to first-time attendees at 2017 and 2020 Ecological Society of America (ESA) meeting

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INVITED SEMINARS

- Panel member on the importance of research in the regenerative agriculture and food system movement. April 29th, 2026. Why Regenerative NYC. Brooklyn Navy Yard.
- *"Detecting change in ecosystem processes: Scale, extent and resolution of empirical data collection".* July 16th, 2025. Cary Institute for Ecosystem Studies. Millbrook NY.
- *"Fluxbots in the wild: Using inexpensive technology to expand the scope of biogeochemical research."* May 13, 2025. Woodwell Climate Research Center, Falmouth MA.
- *"The role of data resolution and extent in detecting the effects of change on ecosystem processes – case studies from methods development in zoogeochemistry and agroecology".* April 21, 2025. Columbia Lamont-Doherty Biology and Paleo Environment division seminar series. Columbia University, Palisades NY.
- *"Hudson Carbon: A place-based research hub in the Hudson Valley."* May 14, 2025. Stone Barns Center for Food and Agriculture, Pocantico Hills NY.
- *"The intersections of basic and applied research: Exploring moose impacts on boreal forest carbon dynamics to optimize natural carbon capture potential".* February 3, 2023. Memorial University, St. John's Newfoundland, Canada.
- "*Wild robots: Using DIY technology to measure carbon flux in a long-term large herbivore experiment*." November 6, 2019. UC Santa Barbara, Santa Barbara CA. Biogeosciences seminar.